

5.4 Word2Vec Skip-gram

Predict the neighbors from the middle word.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. Word2Vec as a prediction game

Word2Vec takes a large unlabeled corpus and puts every word in one vector space. The space is defined by **context**: words that show up in the same windows should end up nearby.

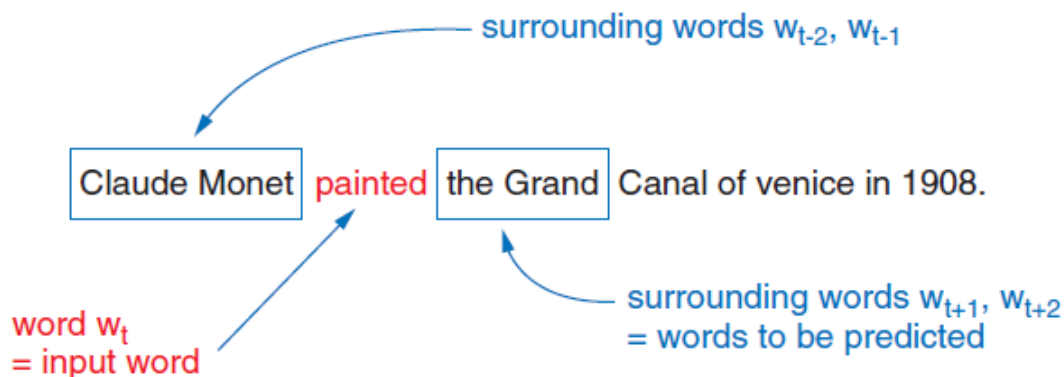
The model starts with random vectors. It refines them by playing a supervised game on made-up labels: given a word, what else stood next to it? A **two-layer** net does the prediction. When training is done, you keep the hidden weights.

Skip-gram is the version of that game that predicts **context from the center**. CBOW (note 5.3) predicts the center from the context.

Sources: *Practical Natural Language Processing* and *Natural Language Processing in Action*.

2. What “skip” means

A skip-gram is an n-gram that is allowed to **skip** tokens. You predict *Claude* from *painted* and skip *Monet*. The window still bounds how far you look.



One center word produces **several** training pairs, one per neighbor. That is why skip-gram “sees” more examples than CBOW on the same sentence.

3. Softmax

The output layer is a classifier over the whole vocabulary. **Softmax** turns raw scores z into probabilities that sum to 1:

$$\sigma(z)_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}}$$

Entry j is "probability that word j is a neighbor of the input." During training you compare that distribution to a one-hot of the true neighbor (cross-entropy). At the end of an update you can pretend the argmax is a one-hot prediction; the useful object is still the **weight matrices**, not those predictions.

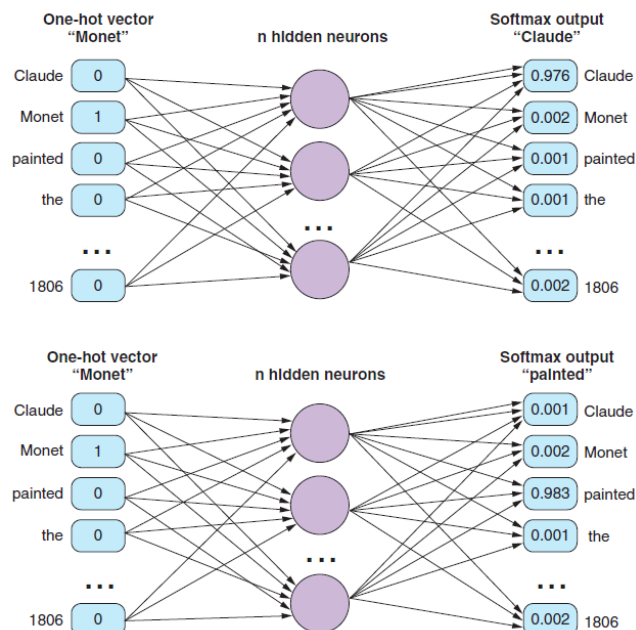
A full softmax over a 100,000-word vocabulary is slow. Note 5.5 replaces it with negative sampling.

4. From a sentence to pairs

Window size 2, same sentence as CBOW:

```
sentence = "Claude Monet painted the Grand Canal of Venice in 1806."
```

A 5-gram tokenizer puts each word in the center once. For four neighbors you get four updates per center: each neighbor is a separate target.



Center	Neighbors you predict (window 2)
painted	Claude, Monet, the, Grand
Canal	the, Grand, of, Venice

Input is the **one-hot** of the center (size $|V|$). Hidden layer is size d . Output is $|V|$ softmax units. Four neighbors $\Rightarrow$ four forward/backward passes with the same input one-hot and four different target one-hots.

After training, **each row of the input-hidden weight matrix** is a word vector. The softmax head is discarded. Those rows are what you look up at inference.

5. CBOW versus skip-gram, in one table

	CBOW	Skip-gram
Direction	context $\rightarrow$ center	center $\rightarrow$ context
Input	pooled / multi-hot context	one-hot center
Examples per window	one	one per neighbor
Tends to win on	frequent words, speed	rare words, small corpora

Both learn the same kind of object: a d -dimensional vector per type. Neither models order inside the window. Neither is a language model you ship. They are **representation** methods.

Count the pairs. Ten tokens and window 2 give about 10×4 skip-gram pairs (edges of the sentence have fewer neighbors). CBOW gives about 10 examples. That factor of ~ 4 is the “more examples” claim, not a different definition of meaning.

6. Practice

1. For center *Monet* and window 2, list the skip-gram training pairs from the sentence above.
2. Why does skip-gram produce more gradient updates per sentence than CBOW?
3. Softmax outputs a distribution over V . After training, do you use that distribution as the embedding? What do you use instead?